

THE MASTER DIAGRAM AND UNRAMIFIEDNESS OF $\mathcal{P}^{[d]_G}$ AT η_C

1. SETUP AND NOTATION

Let $\mathcal{P} \rightarrow U$ be a G -torsor over $U = C \setminus \mathfrak{m}$, where G is a finite abelian group, $H \trianglelefteq G$, and C is a smooth projective curve. Fix $p_\infty \in \mathfrak{m}$ and an integer d bounding the ramification of $\mathcal{P} \rightarrow U$ at p_∞ .

Recall the four quotient schemes

$$\mathcal{P}^{[d]_H}, \quad \mathcal{P}^{[d]_G}, \quad (\mathcal{P}_{G/H})^{(d)}, \quad (\mathcal{P}_{G/H})^{[d]},$$

together with the canonical Cartesian square

$$\begin{array}{ccc} \mathcal{P}^{[d]_H} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{(d)} \\ \downarrow & & \downarrow q \\ \mathcal{P}^{[d]_G} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{[d]} \\ & \searrow G & \downarrow G/H \\ & & U^{(d)}, \end{array}$$

in which the arrows labelled H , G/H and G are torsors under the indicated finite constant groups (in particular, *étale*), and the upper square is Cartesian (prop:torsor_{lower}*diagram*).

The G/H -torsor $\mathcal{P}_{G/H} \rightarrow U$ compactifies to a finite map of smooth projective curves $f: C' \rightarrow C$. Pick $p'_\infty \in f^{-1}(p_\infty)$ with ramification index $e \leq d$. Write

$$f^{(d)}: C'^{(d)} \rightarrow C^{(d)}$$

for the induced map of d -fold symmetric powers; it sends $d \cdot p'_\infty \mapsto d \cdot p_\infty$. Let

$$\pi_C: X_C \rightarrow C^{(d)}, \quad \pi_{C'}: X_{C'} \rightarrow C'^{(d)}$$

be the blowups at $d \cdot p_\infty$ and $d \cdot p'_\infty$ respectively, with exceptional divisors $E_C, E_{C'}$ and generic points $\eta_C, \eta_{C'}$. The DVRs of these generic points are

$$V_C := \mathcal{O}_{X_C, \eta_C} \subset K(C^{(d)}), \quad V_{C'} := \mathcal{O}_{X_{C'}, \eta_{C'}} \subset K(C'^{(d)}).$$

The local relation $t = u \cdot s^e$ between uniformizers at p_∞, p'_∞ propagates to symmetric powers, giving

$$v_{C'}(x) = e \cdot v_C(x) \quad \text{for all } x \in K(C^{(d)})^\times,$$

i.e. $V_{C'}/V_C$ is an extension of DVRs of ramification index exactly e .

2. THE MASTER DIAGRAM

The torsor tower over the open base $U^{(d)}$ embeds into the projective geometry of $C^{(d)}$ and $C'^{(d)}$, which in turn admit canonical blowups at the points at infinity. We assemble all of this into a single commutative diagram.

$$\begin{array}{ccccccc}
 \mathcal{P}^{[d]_H} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{(d)} & \xhookrightarrow{\iota'} & C'^{(d)} & \xleftarrow{\pi_{C'}} & X_{C'} \\
 \downarrow & & \downarrow q & & \downarrow f^{(d)} & & \downarrow f_X \\
 \mathcal{P}^{[d]_G} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{[d]} & & & & \\
 & \searrow G & \downarrow G/H & & & & \\
 & & U^{(d)} & \xhookrightarrow{\iota} & C^{(d)} & \xleftarrow{\pi_C} & X_C
 \end{array}$$

Reading the diagram:

- The **left block** (columns 1–2) is the torsor tower of $\text{prop: torsor_tower_diagram} : H\text{-torsor} \text{ horizontal } U^{(d)}$.
- The **middle block** (column 3) records the compactifications. The map ι is the canonical open immersion $U^{(d)} \hookrightarrow C^{(d)}$. The map ι' is the open immersion $(\mathcal{P}_{G/H})^{(d)} \hookrightarrow C'^{(d)}$ obtained from compactifying $\mathcal{P}_{G/H} \subset C'$. The vertical $f^{(d)}$ is the finite morphism of symmetric powers; restricting along ι' recovers the canonical map $(\mathcal{P}_{G/H})^{(d)} \rightarrow U^{(d)}$, which is the composite of q with the G/H -torsor.
- The **right block** (column 4) records the blowups at the diagonal points at infinity. The dashed arrow $f_X : X_{C'} \dashrightarrow X_C$ is the rational lift of $f^{(d)}$, defined on the strict transform; on generic points of the exceptional divisors it sends $\eta_{C'} \mapsto \eta_C$ and the induced extension of DVRs $V_C \hookrightarrow V_{C'}$ has ramification index e .

3. UNRAMIFIEDNESS OF $\mathcal{P}^{[d]G}$ AT η_C

Theorem 1. *Suppose*

(H1) $(\mathcal{P}_{G/H})^{[d]} \rightarrow U^{(d)}$ *is unramified at η_C ; and*

(H2) $\mathcal{P}^{[d]_H} \rightarrow (\mathcal{P}_{G/H})^{(d)}$ *is unramified at $\eta_{C'}$.*

Then $\mathcal{P}^{[d]G} \rightarrow U^{(d)}$ is unramified at η_C .

Proof. By the tower

$$\mathcal{P}^{[d]G} \xrightarrow{H} (\mathcal{P}_{G/H})^{[d]} \xrightarrow{G/H} U^{(d)},$$

unramifiedness of the composite at η_C follows from unramifiedness of each factor at the corresponding point. The lower factor is unramified at η_C by hypothesis (H1). It therefore suffices to prove the following.

Claim. The H -torsor $\mathcal{P}^{[d]G} \rightarrow (\mathcal{P}_{G/H})^{[d]}$ extends to an unramified cover at any point ξ of $(\mathcal{P}_{G/H})^{[d]}$ lying above η_C .

Fix such a ξ . Pass to local rings:

$$A := \mathcal{O}_{(\mathcal{P}_{G/H})^{[d]}, \xi}, \quad B := \mathcal{O}_{\mathcal{P}^{[d]G}}(\text{base change}),$$

where, more precisely, B is the semilocal ring obtained from $\mathcal{P}^{[d]G} \times_{(\mathcal{P}_{G/H})^{[d]}} \text{Spec } A$. Since $\mathcal{P}^{[d]G} \rightarrow (\mathcal{P}_{G/H})^{[d]}$ is finite, $A \rightarrow B$ is finite, and we must show that it is étale at every maximal ideal of B .

By (H1), the local ring A is a DVR (it is regular of dimension one, with residue field a finite separable extension of the residue field of V_C , by definition of unramifiedness at η_C). The point $\eta_{C'}$ of $C'^{(d)}$ lies above η_C via $f^{(d)}$, and via the open immersion ι' it lies in $(\mathcal{P}_{G/H})^{(d)}$. Let ξ' be the unique point of $(\mathcal{P}_{G/H})^{(d)}$ above ξ corresponding to $\eta_{C'}$ (so that ξ' maps to ξ under q and to η_C under the composite to $U^{(d)}$). Set

$$A' := \mathcal{O}_{(\mathcal{P}_{G/H})^{(d)}, \xi'}, \quad B' := \mathcal{O}_{\mathcal{P}^{[d]H}}(\text{base change to } A').$$

The Cartesian square in `prop:torsor_tower_diagram_localizes_to_a_Cartesian_square_of_finite_ring_extensions`

$$\begin{array}{c} B \\ \uparrow \\ A \end{array}$$

$B \otimes_A A'$.

Step 1: $A \rightarrow A'$ is faithfully flat. Both A and A' are DVRs (the latter by construction, since ξ' specializes to the generic point of the exceptional divisor of $X_{C'} \rightarrow C'^{(d)}$, which is regular of codimension one). The map $A \rightarrow A'$ is the localization of the finite morphism q at the chosen points, hence it is finite. A finite, injective, local homomorphism of DVRs is automatically flat (free of rank equal to the degree of the residue extension times the ramification index), and is faithfully flat because the maximal ideal of A generates a non-unit ideal in A' .

Step 2: $A' \rightarrow B'$ is étale. The cover $\mathcal{P}^{[d]H} \rightarrow (\mathcal{P}_{G/H})^{(d)}$ is an H -torsor over the open base $U^{(d)}$ pulled back along q , hence étale wherever it is defined as such; by hypothesis (H2) it is moreover unramified at $\eta_{C'}$, hence étale at ξ' (a finite, unramified, flat morphism is étale). Therefore $A' \rightarrow B'$ is étale at every maximal ideal of B' .

Step 3: Étale descent. We have the Cartesian square

$$B' = B \otimes_A A',$$

with $A \rightarrow A'$ faithfully flat (Step 1) and $A' \rightarrow B'$ étale (Step 2). By fpqc descent of étale morphisms ([?, <https://stacks.math.columbia.edu/tag/02L4>]; equivalently, [?, Exp. IX, Prop. 4.1]), étaleness descends along faithfully flat morphisms: if $A' \rightarrow B \otimes_A A'$ is étale and $A \rightarrow A'$ is faithfully flat, then $A \rightarrow B$ is étale.

Therefore $A \rightarrow B$ is étale, i.e. $\mathcal{P}^{[d]G} \rightarrow (\mathcal{P}_{G/H})^{[d]}$ is étale at ξ . Since ξ was an arbitrary point above η_C , we conclude that $\mathcal{P}^{[d]G} \rightarrow (\mathcal{P}_{G/H})^{[d]}$ is unramified above η_C .

Combining with (H1), the composite $\mathcal{P}^{[d]G} \rightarrow U^{(d)}$ is unramified at η_C , which proves the claim. $\square$

Remark 1. The proof is purely descent-theoretic: the only properties of the symmetric powers and blowups used are that the relevant local rings A and A' are DVRs, that q is finite, and that the Cartesian square in `prop:torsor_tower_diagram_remains_Cartesian_after_localization`. In particular, no style hypothesis on tameness or coprimality of f with the residue characteristic is needed.

Remark 2 (Why descent and not multiplicativity). Naive multiplicativity of ramification indices in the diamond

$$V_C \subset V_1 \subset W_1 \subset W_2, \quad V_C \subset V_1 \subset V_2 \subset W_2$$

gives only $e_1 \cdot e(W_2/W_1) = e$, where $e_1 = e(W_1/V_1)$ is the ramification of $\mathcal{P}^{[d]G}/(\mathcal{P}_{G/H})^{[d]}$ at ξ that we wish to control. Without further input this constrains e_1 only by $e_1 \mid e$ (in the tame range, by Abhyankar's lemma). The Cartesian-ness of the upper square is precisely what upgrades this to the strict equality $e_1 = 1$: it identifies $B' = B \otimes_A A'$, which is the hypothesis of fpqc descent of étaleness.

REFERENCES

- [1] A. Grothendieck, *Revêtements Étales et Groupe Fondamental (SGA 1)*, LNM 224, Springer, 1971.
- [2] The Stacks Project Authors, *Stacks Project*, <https://stacks.math.columbia.edu>.